

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Computer Systems, Computer Networks, Digital Design & Computer Architecture, Systems Security, Cybersecurity, Algorithms & Data Structures, Object-Oriented Design, Foundations of Software Engineering, Foundations of Data Science

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own the front-of-house device stack for 4 Wayfair retail brands: e-labels, mobile, and in-store POS
- Building a supplier-agnostic e-label API unifying 4 hardware vendors behind one interface, deployed to new outlets
- Shipped POS checkout flows to production: scan-to-basket customer switching and live associate-entry sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Overhauled Relay JMS broker serialization with json-io across 20+ distributed services, eliminating timeouts
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing production workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Held 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Continue as technical advisor on architecture and agentic-workflow direction

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Built CI/CD tooling with Azure DevOps and K8s, enabling rapid, reliable deployment and rollback

PROJECTS

Kepler | Rust, C++, Raft, LSM-tree, MVCC, Tokio

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and storage engine
- Implemented Raft leader election, log replication, and membership changes over async Tokio TCP transport
- Added MVCC transaction layer giving snapshot isolation and linearizable reads across a multi-node cluster
- Built memtable, WAL, and leveled SSTable storage with single-tier compaction rewriting every 4 flushes

NOAA AUV Sonar Detection | Python, PyTorch, AWS Neuron SDK, SageMaker

April 2025 – July 2025

- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits
- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Tuned architecture and loss formulation against class imbalance and sensor noise, improving detection

Sentinel | Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker

Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy

SKILLS & HONORS

Languages/Tools: C, C++, Python, Rust, Java, Bash, Linux, TypeScript, JavaScript, Groovy, Docker, Kubernetes, AWS, Azure, Terraform, Jenkins, Buildkite, PostgreSQL, GraphQL

Frameworks & Libraries: FastAPI, Flask, PyTorch, TensorFlow, Pandas, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Tech Lead at Northeastern Electronic Racing, Dean's List (All Semesters), json-io OSS Contributor, PawHacks